Kinetic theory of gases: Maxwell-Boltzmann distribution, Mean free path, diffusion and viscosity

May 25, 2026

Much of the material in this note is based on chapter 1 of David Tong's lecture notes on Kinetic Theory. The figures are also adapted from there.
Boxed text corresponds to material that goes beyond what was covered in the lecture.

1 Velocity distribution in an ideal gas

An ideal gas is a system whose molecules do not interact with each other. This means that (in the absence of any external potential, such as a gravitational field, which we will ignore), the total energy of the gas is simple the sum of all the kinetic energies of the particles, $E = \sum_i \frac{m|\vec{v}_i|^2}{2}$. While this is an idealization, ideal gases give a good first approximation to the properties of many real gases. The quality of this approximation improves as the gas gets more dilute (i.e., with decreasing density N/V).

In the canonical ensemble at temperature T, the probability of a configuration with energy E is proportional to the Boltzmann factor $e^{-\frac{E}{k_B T}}$ Let's say that we are interested in some quantity $g(\vec{v}_i)$ that only depends on the velocity of the i-th particle, $\vec{v}_i$. An example might be the speed $g(\vec{v}_i) = |\vec{v}_i|$. Now, we can calculate the average of this quantity in the canonical ensemble. For an ideal gas, we find

$$\overline{g(\vec{v}_i)} = \frac{\int \mathrm{d}^3x_1 \ldots \int \mathrm{d}^3x_N \int \mathrm{d}^3v_1 \ldots \int \mathrm{d}^3v_N e^{-\frac{E}{k_B T}} g(v_i)}{\int \mathrm{d}^3x_1 \ldots \int \mathrm{d}^3x_N \int \mathrm{d}^3v_1 \ldots \int \mathrm{d}^3v_N e^{-\frac{E}{k_B T}}} = \frac{\int \mathrm{d}^3v_i e^{-\frac{m|\vec{v}_i|^2}{2k_B T}} g(\vec{v}_i)}{\int \mathrm{d}^3v_i e^{-\frac{m|\vec{v}_i|^2}{2k_B T}}} = \int \mathrm{d}^3v_i p_{1\mathrm{p}}(\vec{v}_i) g(\vec{v}_i). \tag{1.1}$$

What we see is that for such quantities, that depend only on the properties of just a single particle, we can replace the full probability distribution with a one-particle distribution $p_{1\mathrm{p}}(\vec{v})$. In the present case, where all particles are equivalent, these one-particle distributions are the same for all of them and it reads

$$p_{1\mathrm{p}}(\vec{v}) = \left(\frac{m}{2\pi k_B T}\right)^{3/2} e^{-\frac{m|\vec{v}_i|^2}{2k_B T}}. \tag{1.2}$$

We can go further if we assume that the quantity $g(\vec{v})$ depends on the on the magnitude $|\vec{v}| = v$ of the velocity. In this case, we can write $\vec{v}$ in polar coordinates and integrate over its direction to get $\overline{g(v)} = \int \mathrm{d}v f_{\mathrm{MB}}(v) g(v)$, where f_{MB} is the so-called *Maxwell-Boltzmann distribution*, expressing the probability of finding a particle with speed close to v. It is given by

$$f_{\mathrm{MB}}(v) = \left(\frac{m}{2\pi k_B T}\right)^{3/2} 4\pi v^2 e^{-\frac{mv^2}{2k_B T}}, \tag{1.3}$$

where the $4\pi v^2$ is the surface area of a sphere with radius v. While the derivation relied on having an ideal gas, distribution of speeds can be measured experimentally in real gases and is found to obey Eq. (1.3) to a good precision.

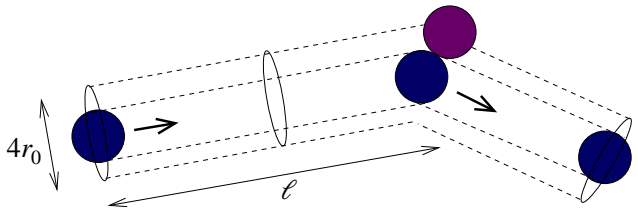


Figure 1: Particle collisions and mean free path. We can model gas molecules as hard sphere of radius r_0. The travel a distance ℓ before colliding with another molecule.

2 Particle collisions and mean free path

Despite its success in describing the velocity distribution in real gases, there is something strange about the Maxwell-Boltzmann distribution. To see this, let us calculate the average speed of a gas molecule. We get $\bar{v} = \int \mathrm{d}v f_{\mathrm{MB}}(v) v = \sqrt{\frac{8k_{\mathrm{B}}T}{\pi m}}$. For a typical gas at room temperature we get $\bar{v} = 10^2$ - 10^3 $\frac{\mathrm{m}}{\mathrm{s}}$ (e.g., for nitrogen molecules in air, it is $\approx 422\frac{\mathrm{m}}{\mathrm{s}}$). A molecule at this speed would cross a room at a fraction of a second. Yet, we know that if someone say were to suddenly open a box full or rotten vegetables in a large room, it would take a considerable amount of time before we could smell it at the other end of the room. How to reconcile these facts?

The answer is that, while $\bar{v}$ gives a fairly good approximation of the speed of a gas molecule at any given instant of time, the molecule will not travel with the same velocity in the same direction for a very long time. Instead, it will collide with other gas particles and these collisions will lead to a change in the direction of its velocity.

We can estimate the time between two collisions using a simple toy model. Let us approximate each molecule as a hard ball of radius r_0. As the particle moves in s straight line between collisions, we can draw a cylinder of radius $2r_0$ around it: if there is another particle whose center is within this cylinder, then the two will collide; see Fig. 1. If the particle travels at speed $\bar{v}$ for a time t, the volume of the corresponding cylinder is $4\pi r_0^2 t\bar{v}$. We get the number of collisions within this time window by multiplying this volume with the density $\rho = V/N$ of the gas[1]. This means that the number of collisions per unit time is

$$\frac{1}{\tau} := \frac{N_{\mathrm{coll}}}{t} = 4\pi r_0^2 \bar{v}\rho. \tag{2.1}$$

We used this formula to define the *scattering time* τ, the average time between that passes between two subsequent collisions. Using the numbers $\rho \approx 10^{19}\frac{1}{\mathrm{cm}^3}$, $\bar{v} \approx 10^5\frac{\mathrm{cm}}{\mathrm{s}}$ and[2] $r_0 \approx 10^{-8}$cm we get that $\tau \approx 10^{-9}$s, meaning that the molecule will only travel for a short time before it collides with another one.

The distance the particle travels between two collisions, called the *mean free path*, is $\ell = \bar{v}\tau \approx 10^{-4}$cm. This is very short compared to macroscopic distances, like the length of a room. This explains our earlier conundrum about why smells do not spread very fast. On the other hand, ℓ is much larger than the size of the molecule r_0 itself. This means that the gas is *dilute*, particles are typically very far from each and most of the time they do not interact. This low density justifies many approximations used in discussing gases, like the van der Waals equation of state.

[1] In fact, instead of $\bar{v}$ we should use the typical *relative* speed $\bar{v}_{\mathrm{rel}}$ between two particles. Assuming the velocities of the two particles are independent, this can be calculated from the Maxwell-Boltzmann distribution. The answer is $\bar{v}_{\mathrm{rel}} = \sqrt{2\overline{v^2}} = \frac{\sqrt{3\pi}}{2}\bar{v}$.. Given that this is a simple heuristic calculation, we will ignore this difference.

[2] Of course molecules are not hard balls. Nevertheless, we there exists a characteristic distance between molecules, which can be measured experimentally, at which they start to strongly repel each other.

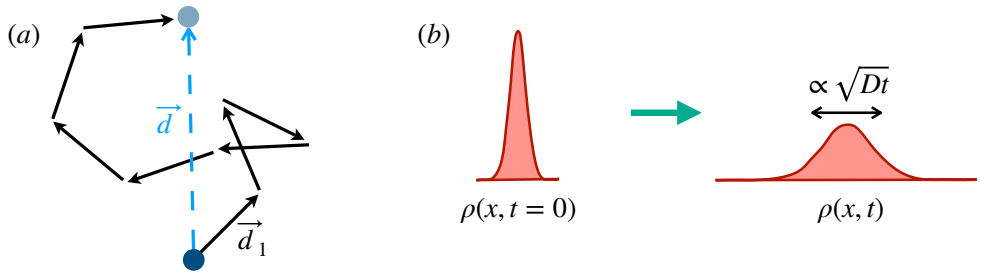


Figure 2: Diffusion of particles. (a) A single particle moves randomly due to its collisions with other particles. Its total displacement $\vec{d}$ is a sum of many such random displacement $\vec{d}_i$. (b) As a result, the density of particles $\rho(x, t)$ evolves according to the diffusion equation (3.2). By time t, a "bump" in the density spreads out within a region of width $\sqrt{Dt}$.

3 Diffusive motion of particles

We found that the gas particles suffer collisions at a high rate. What is the overall trajectory of a molecule after many such collisions?

Let $\vec{d}_i$ denote the displacement vector of a particle between its i-th and $(i+1)$-th collisions. The overall displacement of the particle is the sum of all these vectors, $\vec{d} = \sum_i \vec{d}_i$ (see Fig. 2(a)) In reality, all the $\vec{d}_i$ vectors have different lengths, since the particle's speed and the times between collisions are random. However, we will simplify things by neglecting this randomness and taking all of them to have lengths equal to the mean free path, $|\vec{d}_i| = \ell$. We will also take the time between any two subsequent collisions to be exactly equal to the scattering time τ.

We make another important simplifying assumption: we will assume that each collision makes the direction of the particle's velocity completely random, erasing all memory of which way it was going before the collision. This means that the average displacement between collisions is zero, $\overline{\vec{d}_i} = 0$, since all possible directions are equally probable. Of course this does not mean that the particle is staying in place. We can instead calculate the *root mean square displacement* to find

$$d_{\text{rms}}^2 := \overline{d^2} = \overline{\left|\sum_i \vec{d}_i\right|^2} = \overline{\sum_{i,j} \vec{d}_i \cdot \vec{d}_j} = \sum_{i,j} \delta_{i,j} \ell^2 = \sum_i \ell^2. \tag{3.1}$$

Here, we used that, since the different displacement vectors point in independently random directions, their scalar product, $\vec{d}_i \cdot \vec{d}_j$ averages to zero, unless $i = j$, in which case it equals $|\vec{d}_i|^2 = \ell^2$.

In a time interval t, the number of collisions is t/τ, so we get the final result that the mean square displacement behaves as

$$d_{\text{rms}} = \sqrt{\frac{\ell^2}{\tau} t} = \sqrt{Dt}, \tag{3.2}$$

where $D = \ell^2/\tau$. What we see is that the frequent collisions lead to qualitatively new behavior: the displacement grows as a square root of the time elapsed. Such behavior is called *diffusive* and is characteristic of random ("stochastic") motion, both in physics and other fields[3]. It should be contrasted with the more conventional *ballistic* motion of a projectile shot from a gun, which travels in one direction and covers a distance growing linearly with the time elapsed. Instead of a velocity, the speed of diffusive motion is characterized by its *diffusion constant* D.

There is another, more macroscopic manifestation of the diffusive motion of particles. Rather than focusing on a single gas molecule, let us ask how inhomogeneities in the density of the gas evolve over time. So far, we encountered the average density, $\rho = N/V$. However, in practice, the density might not be uniform in space, certain regions of the gas might contain a surplus of particles while others contain

[3]In physics, one context where such diffusive motion was first understood is that of so-called *Brownian motion*.

less. We can describe these using a function $\rho(\vec{x})$ for the density at position $\vec{x}$[4]. As particles move around, the function $\rho(\vec{x})$ will also change in time. This change also reflects the diffusive motion of the particles.

To see how diffusion manifests macroscopically, imagine a situation where the density is the same almost everywhere, except at one particular location, where it is higher than everywhere else. As time passes, the particles spread out from their initial positions and this "extra" density also spreads out and slowly melts into the background. Since the particles move diffusively, as in Eq. (3.2), by time t, it spreads out into a region of radius $r(t) = \sqrt{Dt}$; at the same time, the difference in density in this region compared to the background goes down, since the total number of particles, $\int \rho(\vec{x}) \mathrm{d}^3x$ does not change. This process is illustrated in one dimension in Fig. 2(b).

One can describe this evolution of the density using the *diffusion equation*. In the one-dimensional case it describes the evolution of the density as a function of the time t and the coordinate x as

$$\partial_t \rho(x,t) = D\partial_x^2 \rho(x,t). \tag{3.3}$$

In three dimension, we can write similar equation but with ∂_x^2 replaced by the Laplace operator $\vec{\nabla}^2$. A similar diffusion equation can be derived for the spreading of energy in the system, where the $\rho(\vec{x})$ is replaced by the energy density at position $\vec{x}$.

One can derive the diffusion equation from the microscopic picture of diffusive motion. We illustrate here the idea in one dimension. Consider how the density at location x changes in a small time interval Δt. It changes due to particles from other locations moving to position x. These other particles arrive from some distance Δx away, which is itself a random variable. So we can formally write it as

$$\rho(x, t+\Delta t) = \overline{\rho(x-\Delta x, t)}, \tag{3.4}$$

where the overline denotes averaging over the distances Δx that the particles travel to get to x. We assume that these distances (which are on the order of the mean free path ℓ) are small compared to the scale on which the density is defined (which must be much bigger than the typical inter-particle distance), We can then expand the right hand side of Eq. (3.4) to get

$$\overline{\rho(x-\Delta x,t)} \approx \rho(x,t) - \overline{\Delta x}\,\partial_x\rho(x,t) + \frac{1}{2}\overline{(\Delta x)^2}\,\partial_x^2\rho(x,t) = \rho(x,t) + \frac{D}{2}\Delta t\,\partial_x^2\rho(x,t), \tag{3.5}$$

where we used Eq. (3.2) to write $\overline{(\Delta x)^2} = D\Delta t$ and that $\overline{\Delta x} = 0$. Combining Eq. (3.4) with Eq. (3.5) and taking $\Delta t \to 0$ gives the diffusion equation $\partial_t \rho = \frac{D}{2}\partial_x^2\rho$. This is almost the same as in Eq. (3.3), except for a factor of $1/2$. In fact, it is more conventional to define D such that Eq. (3.2) holds without the additional factor of $1/2$, so we should redefine the diffusion constant such that $\overline{(\Delta x)^2} = \sqrt{2Dt}$. A similar calculation shows that in three dimensions we should define it as $\overline{|\Delta\vec{x}|^2} = \sqrt{6Dt}$ to get the diffusion equation $\partial_t\rho = D\vec{\nabla}^2\rho$.

4 Viscosity

Another macroscopic property of fluids which can be understood from the microscopic perspective of particle collisions is *viscosity*. This derivation leads to the surprising conclusion that the viscosity is independent of the density of a fluid. While this claim is counter-intuitive, and it greatly surprised Maxwell when he first derived it, it turns out to be experimentally correct to a good approximation[5]

Let us imagine a fluid in a container of depth d, with the top surface of the fluid moving at a constant speed u relative to the bottom which we take to be at rest[6]. Let z denote the vertical coordinate (we

[4]To make sense of $\rho(\vec{x})$, we should imagine dividing the container that contains the gas into many smaller regions (so-called *fluid cells*) in such a way that the size of each cell is small compared to that of the whole system but it is large enough that it contains a macroscopically large number of particles. $\rho(\vec{x})$ is then the density of a cell located near $\vec{x}$. These fluid cells should not be confused with the phase space cells that appear in the formulation of statistical physics. The latter are regions in *phase space* while the fluid cells are regions in physical space (e.g. a particular corner of a room).

[5]One example of this can be observed quite easily. Oil (e.g. olive oil or sunflower oil) is less dense than water, demonstrated by the fact that it floats on top of water. Nevertheless, it is also quite clearly more viscous.

[6]Note that viscosity is usually relevant in the context of a *liquid*, rather than a gas. There is no fundamental difference between liquids and gases, they differ only in how closely packed the particles are: a liquid is much less dilute than a gas is. As a result, the same basic picture of particle collisions also applies to liquids.

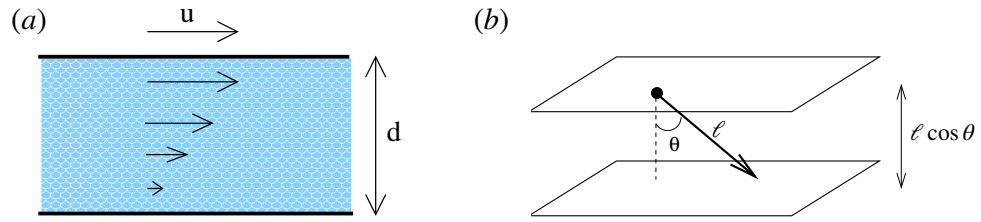


Figure 3: Calculating the viscosity. (a) Shows the setup with a fluid flowing in the x direction at a speed $u(z)$ that depends on its z coordinate. (b) Shows a particle traveling between two different layers of the fluid between two collisions.

take x to be the direction in which the fluid is flowing). We divide the fluid into small layers of width $\mathrm{d}z$. The layer at height z moves as speed $u(z)$ where $u(z)$ interpolates between $u(0) = 0$ and $u(d) = u$. The frictional force F between two subsequent layers is proprtional to the surface area A of the fluid and, assuming $u(z)$ changes slowly, it is also proportional to the speed difference between the layers. The viscosity η is defined as the proportionality constant, such that

$$F = \eta A \frac{\mathrm{d}u(z)}{\mathrm{d}z}. \tag{4.1}$$

To calculate η we need to answer the question: how much does the momentum of a given fluid layer change in one unit of time? Consider a particle moving between two collisions. As we know, it travels a typical distance ℓ before colliding again. However, now during this travel, it will also move to a different layer of the fluid. In particular, its z coordinate will change by $\Delta z = \ell \cos\theta$ where θ is the angle between its velocity vector and the vertical axis. This means that when it next collides, it is with particles that have a different average speed $u(z+\Delta z)$ in the x direction. Assuming, an elastic collision, the two particles exchange their velocities: as a result, the momentum of the second particle changes by an amount

$$\Delta p_{1\text{ collision}} = m(u(z+\Delta z) - u(z)) \approx m\frac{\mathrm{d}u}{\mathrm{d}z}\ell\cos\theta. \tag{4.2}$$

This is the momentum exchanged in a single collision. We next need to determine how many collisions occur in a unit time interval. The particles that reach the layer at coordinate z are those whose velocities have a sufficiently large z component. In particular, the particles reaching the layer in a time Δt originate from a slab of width $v_z\Delta t$, making the number of particles passing through equal $\rho A v_z \Delta t$.

Putting all of this together, we have that the force between layers, which is equal with the rate of change of the layer's total momentum, is given by

$$F = \frac{\mathrm{d}p}{\mathrm{d}t} = \rho A \int \mathrm{d}^3 v p_{1\mathrm{p}}(\vec{v}) v_z \Delta p_{1\text{ collision}}. \tag{4.3}$$

Here $p_{1\mathrm{p}}(\vec{v})$ stands for the distribution over the speeds of a single particle. The collisions between particles mean that they interact, and the fluid is not an ideal gas (indeed, it might even be a liquid), so in principle this distribution is different from the one we saw in Eq. (1.2). Luckily, as we will see, the result we get will not depend on the details of this one-particle distribution.

To evaluate Eq. (4.3), we can write the velocity in polar coordinates as $\vec{v} = \begin{pmatrix} v\sin\theta\cos\phi \\ v\sin\theta\sin\phi \\ v\cos\theta \end{pmatrix}$. In these polar coordinates, the infinitesimal volume reads $\mathrm{d}v^3 = v^2\sin\theta \mathrm{d}v\mathrm{d}\theta\mathrm{d}\phi$. Plugging in Eq. (4.2) for $\Delta p_{1\text{ collision}}$, and comparing with Eq. (1.3), we end up with the following formula for the viscosity:

$$\eta = \rho m \ell \underbrace{\int_0^\infty \mathrm{d}v \frac{f(v)}{4\pi} v}_{\bar{v}/4\pi} \underbrace{\int_0^\pi \mathrm{d}\theta \cos^2\theta \sin\theta}_{2/3} \underbrace{\int_0^{2\pi} \mathrm{d}\phi}_{2\pi} = \frac{m\rho\ell\bar{v}}{3}, \tag{4.4}$$

where $f(v)$ is the probability dsitribution for the magnitude of the velocity of a single particle. Again, this might differ from the Maxwell-Boltzmann distribution in Eq. (1.3), since we are not dealing with an ideal gas. However, the details of the distribution do not matter, we see that whatever the distribution is, the integral over v in Eq. (4.4) gives the average speed of particles, $\bar{v}$

Eq. (4.4) gives the viscosity in terms of microscopic properties of the gap particles. At first glance, it looks like the viscosity should increase with the density, wince it contains a factor of ρ. Remember, however, that the mean free path is given by $\bar{v}\tau$ and by our earlier formula (2.1) for the scattering time τ, this means that $\ell \propto 1/\rho$. Thus, the density-dependence of the mean free path cancels the factor of ρ in Eq. (4.4), making the viscosity (within the approximations we used, at least), independent of the density. The reason for this can be seen from the calculation above: a smaller density means fewer collisions that can impart momentum, but the particles taking part in those collisions tend to come from further away and thus carry more momentum. These two effects cancel each other.